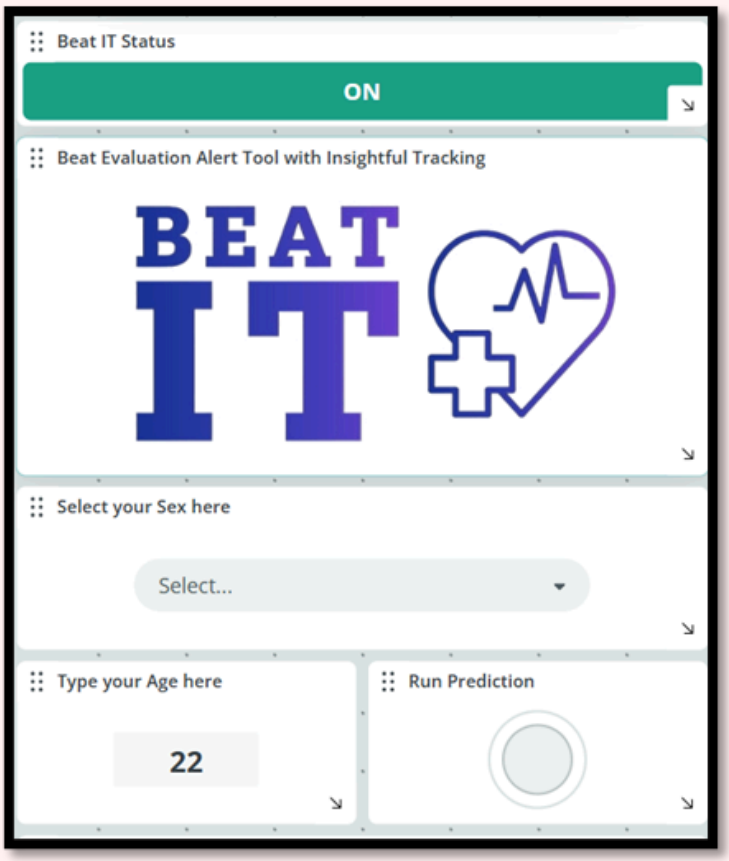
BEAT IT

USER MANUAL

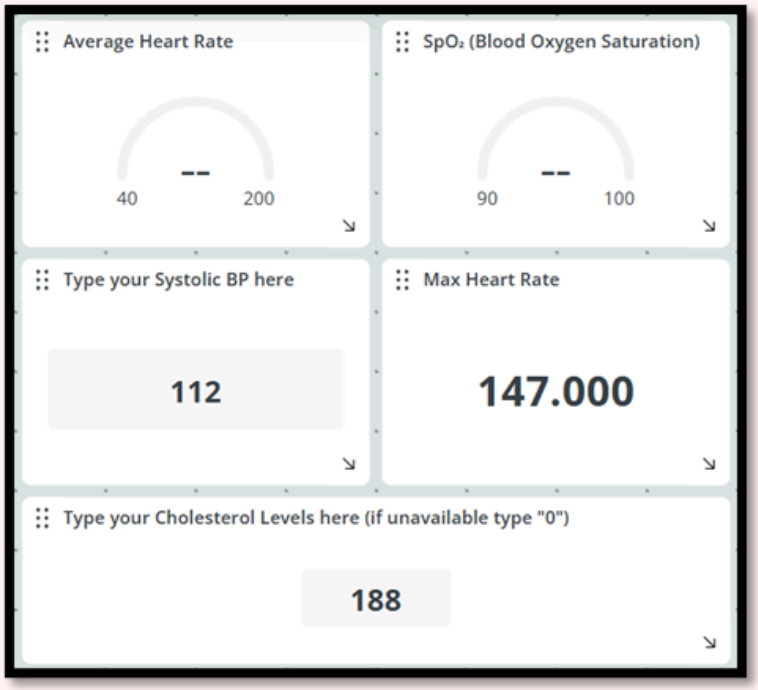
I. SYSTEM OVERVIEW AND INPUT SCREEN

SECTION 1. BEAT IT STATUS & USER INFO

This is the main system control and user demographic input area. It shows the system is **ON**. Users must select their **Sex** and type their **Age** (e.g., 22) to enable risk prediction. The **Run Prediction** button is used to initiate the analysis after all necessary data is entered.



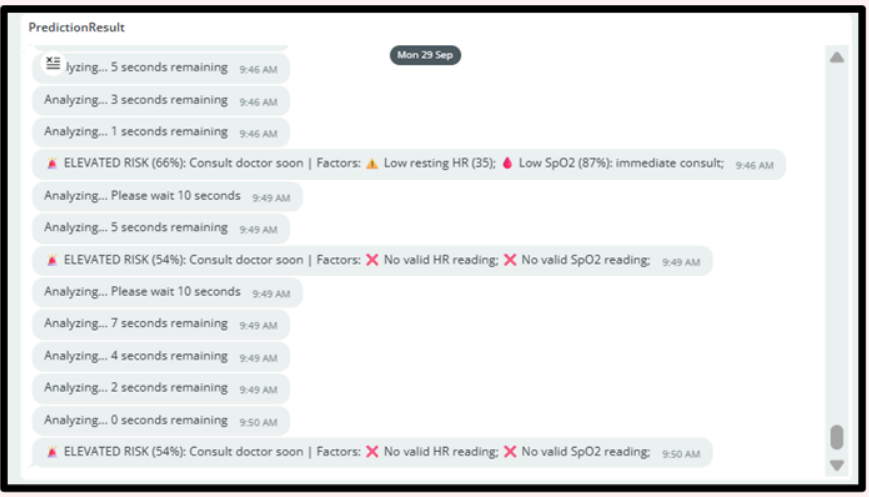
SECTION 2. VITALS INPUT SCREEN



This shows the real-time heart rate monitoring of the user that includes: **Average Heart Rate, Max Heart Rate, and SpO2 (Blood Oxygen Saturation)**. This is also where the user manually enters key health metrics. The screen displays input fields for: **Systolic BP, and Cholesterol Levels**. The ranges for HR (40-200) and SpO2 (90-100) are shown on the gauge displays.

II. PREDICTION RESULTS AND HISTORY

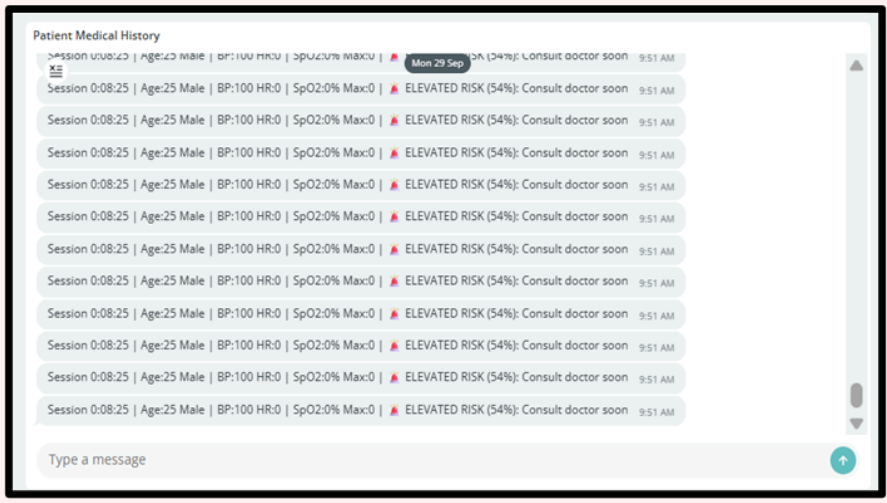
SECTION 3. PREDICTION RESULT LOG



This log displays the results of individual risk assessments. Each entry shows the analysis status (e.g., "Analyzing..."), the final risk level (e.g., **ELEVATED RISK** (66%) or (54%)), and a recommended action ("Consult doctor soon" or "immediate consult"). It also lists the contributing **Factors** (e.g., "Low resting HR (35)," "Low SpO2 (87%)," or "No valid HR/SpO2 reading").

SECTION 4. PATIENT MEDICAL HISTORY

This screen is a chronological record of past monitoring sessions. Each session log details the **Session duration** (e.g., 0:08:25), **Age** (e.g., 25 Male), recorded vitals (**BP**, **HR**, **SpO2**, the calculated **Risk** (e.g., **ELEVATED RISK (54%)**), and the timestamp. This provides a long-term view of the patient's health status.



III. RISK CATEGORIES AND DATA VISUALIZATION

SECTION 5. RISK CATEGORY TABLE

This table is the key for interpreting the percentage risk score. It defines four categories based on the risk percentage (%) and provides the corresponding action/explanation

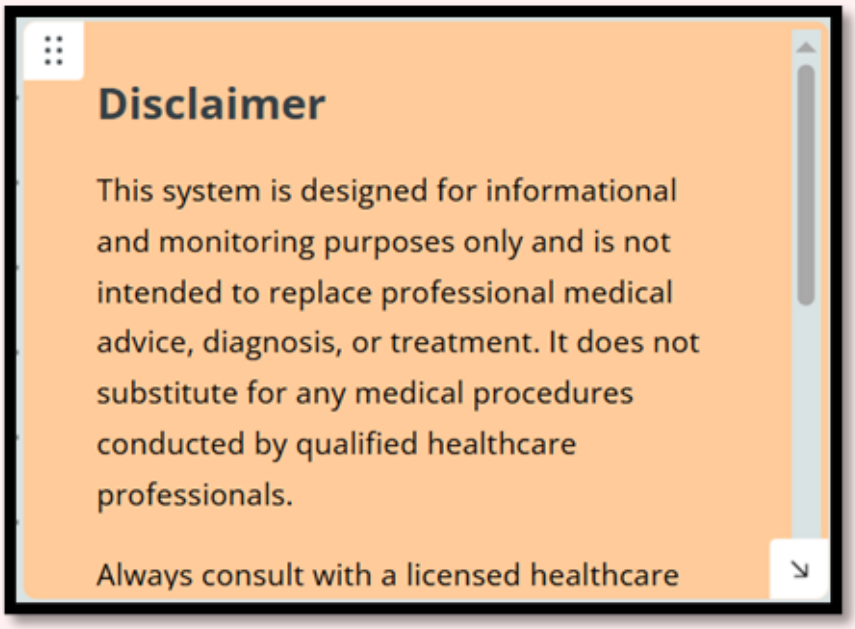
sk Category	Explanation
Low Risk (≤ 20%)	Your heart health appears in a healthy range. Maintain your current lifestyle with regular exercise, balanced diet, and routine health check-ups.
Stage 1 Risk (20–39%)	You have mild risk factors indicating Stage 1 Hypertension or early warning signs. It is advisable to adopt healthier habits, monitor your blood pressure regularly, and consult your doctor for tailored advice.
Stage 2 Risk (40–69%)	You are showing moderate risk with signs of Stage 2 Hypertension. Immediate lifestyle changes and medical consultation are recommended to prevent serious complications.
High Risk (≥ 70%)	Your heart health is at high risk. Seek urgent medical evaluation and treatment to avoid serious cardiovascular events. If you experience symptoms like chest pain or shortness of breath, call emergency services immediately.

SECTION 6. HEART RATE AND SPO2 CHART

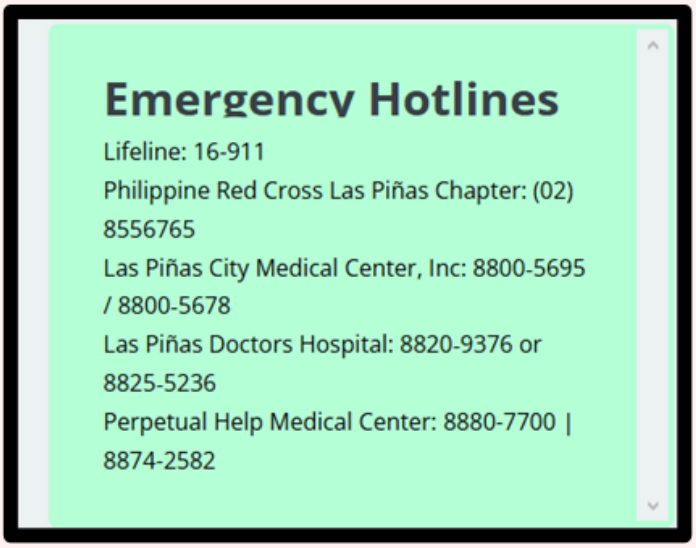
This is a real-time graphical representation of the user's recorded **Average Heart Rate** (red line) and **Average SpO2** (blue line) over time. The user can switch the view between **7 D** (days), **1 D**, **1 H** (hour), and **LIVE** data.

IV. SAFETY AND EMERGENCY INFORMATION

SECTION 7. DISCLAIMER



This is a crucial notification stating that the system is for **informational and monitoring** purposes only. **It is not intended to replace professional medical advice, diagnosis, or treatment.** Users are strongly advised to **always consult with a licensed healthcare professional.**

SECTION 8. EMERGENCY HOTLINES

This provides a list of local emergency and medical contact numbers, specifically for the Las Piñas area, to be used in case of a health crisis.